FACULTY OF COMPUTING

SECD2613 – 04

SYSTEM ANALYSIS AND DESIGN

2025/2026 – SEMESTER 2

PROJECT

Phase 1- Project Proposal and Planning

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1.0 INTRODUCTION

In the high-pressure environment of university life, both students and lecturers are constantly juggling an overwhelming mental load. Between back-to-back lectures, intensive research and the endless cycle of assignments, staying afloat is a challenge. While many still rely on traditional paper planners or basic digital calendars, these methods often crumble under the weight of complex projects and high-stake deadlines. This lack of a cohesive system frequently leads to a fragmented workspace, mounting stress, and the inevitable "human error" that occurs when things get too messy.

The struggle is further complicated by "app fatigue." When essential information is scattered across various platforms, users spend more time switching between tabs than getting work done. This constant digital hopping is more than just a nuisance. It’s a productivity drain that leaves students feeling burnt out and disorganized. Without a centralized hub, balancing academic rigor with extracurricular commitments becomes a losing game, often resulting in missed opportunities and a lingering sense of being overwhelmed.

Communication presents its own set of hurdles. While social media platforms like WhatsApp or Telegram are convenient for quick chats, they are in poor environments seeking help. Critical academic questions often get buried under a mountain of unrelated messages, and finding reliable answers later becomes a frustrating scavenger hunt.

This is exactly why EduHub was developed. It’s designed to pull these moving parts into one intuitive, centralized home. By featuring a specialized Q&A forum where students can post specific hurdles and receive targeted answers, the system acts as a living knowledge base. Combined with comprehensive archives of past papers, recordings of physical lectures and smart tools like an AI-driven progress tracker, UTM Tutoring creates a supportive ecosystem that turns the chaos of university life into a structured path toward success.

2.0 BACKGROUND STUDY

With technological advancement, many institutions of higher education are transforming digital platforms that promote adaptability and learning through data analysis. The number of people at the Universiti Teknologi Malaysia (UTM) comprises many undergraduate students, postgraduate researchers, and academic staff. They are using several platforms and technologies that help them cope with their daily activities. Some examples of the existing platforms include the present UTM e-learning platform, UTMSmart App, MY UTM portal, communication apps such as Telegram and WhatsApp, and physical resources such as textbooks and notes.

These platforms provide individual benefits but cannot function collectively to achieve organisational goals. Students need to navigate from one tool to another to search for lecture notes, deadlines, and other information related to academics, and even ask questions in case they want to clarify some issues about their course material. It is also referred to as "app fatigue" and can negatively affect the mind of the user and reduce overall academic effectiveness. According to a study in educational technology, students who lack access to an integrated learning platform are more likely to fail in exams and miss deadlines.

The accessibility of lectures is one of the concerns. Unlike other universities that are in more advanced environments, UTM does not have an established way of distributing lecture recordings or past year exam papers. If a student misses class due to health issues or any transportation challenges, he is already disadvantaged from the start, since there is no guarantee that there will be a reliable alternative way to access the material. If he wanted to do any revision based on the past year model, there is no guarantee for him getting them.

Furthermore, there is no provision for students to receive feedback in a timely manner regarding their performance in academics. Students will know about their poor academic standing at the end of the semester due to the absence of a centralised system of carry marks tracking. One of the key reasons that contribute to stress among students and lead to poor performance is the absence of pragmatic academic feedback.

Together, all these contribute to making university life unnecessarily difficult for students. It does not mean that the students do not try hard enough or deliver quality work, but it means that there is no sophisticated and integrated system in place to cater to the needs of the UTM community. EduHub can help solve this problem

3.0 PROBLEM STATEMENT

3.1 Fragmented Academic Tools

Normal teaching requires both lecturers and students to deal with several distinct platforms. There is no centralised platform for everything, hence the need for users to switch back and forth among different applications.

3.2 Unavailable Lecture Content

Those students who are unable to participate physically in class lessons due to sickness or any other personal issue have no way of accessing the lesson content, which would otherwise assist them during studies. Currently, there are no uniform guidelines which help in providing recorded lessons in real-time.

3.3 Limited Access to Academic Carry Mark

They do not get real-time and relevant data concerning their academic status and mark progression. It is common for students not to be aware that they are underperforming until it is too late for them to improve their academic situation during the semester.

3.4 Lack of a Centralised Academic Recourses

The past year papers, notes, and other academic material are not made available uniformly in personal computers and even through social circles. There is no institutional archive where such material can be stored and shared equally amongst all the students.

4.0 PROBLEM SOLUTION

A) Classroom Recording Archive

The EduHub application has a feature where lectures held in classrooms can be recorded and saved so that the students have access to the recordings. Students who cannot attend physically can watch the complete recording anytime. In this way, students will not face a disparity because of unavoidable reasons. There will be no need for lecturers to repeat themselves, as they will be able to focus their time on other issues.

B) Academic Forum

There will be an organized Q&A forum with subjects tagged for easy identification of questions raised and answers provided. Questions and answers will always remain indexed and searchable within the platform. Thus, there will be a living institutional memory that is growing rich every semester as far as knowledge acquisition is concerned. Students will get answers instantly while lecturers will be aware of what confuses their learners.

C) Performance Analysis Tracker

An AI study assistant analyses the learning pattern and performances of a user to come up with personalized recommendations. It identifies areas of knowledge gaps and tracks carry marks.

D) Centralised Test Bank Repository

EduHub also houses an extensively curated library of examination papers from previous years as well as academic sources, which are quality-controlled by the platform. All UTM users get the same access to the library material without favouritism or discrimination depending on their connections or background.

E) Student Collaboration

This feature will allow user to connect or chat with seniors or lecturers in real time. This can give them more connection and make more friends while gaining more knowledge overtime. This can give the shy users more courage to breakthrough their shyness on searching more colleagues.

F) Study Guide Section

With some user doesn't have any ideas or starter to study. This feature can give the user to make a new or follow a study guide that a senior or lecturer share with. This feature also will give a many study tools that user can use to study with an example of pomodoro timer where user can upload the reading material and press start button, the timer will start , it is 25 minutes per session , after that, there will be a report , quiz and summary.

4.1 Technical Feasibility

Implementation of EduHub can be done from a technical standpoint by adopting the stack that is both current and feasible for use in a university-based setting. The front-end will be created using React.js while the back end will be created using Node.js and Express. Data will be stored using MySQL while videos and other files will be stored using AWS S3. Artificial intelligence capabilities will be implemented through an independent Python Flask server that utilizes the OpenAI API.

4.2 Operational Feasibility

Adoption and maintenance of EduHub will be effortless. Since it is a web application, there is no need for installing any software; therefore, students and lecturers will simply need to log in using their UTM emails via any of the recent browsers available. The application has been designed based on academic work processes in order to ease its adoption for new users. Furthermore, the application's modular code structure allows developers to make changes in one specific layer of the platform without affecting others. Moreover, cloud-based hosting enables easy scalability according to the number of users, and maintenance will be provided after deployment.

4.3 Economic Feasibility (CBA)

Assumptions	
Discount rate	10%
Sensitivity factor (costs)	1.1
Sensitivity factor (benefits)	0.9
Annual change in production costs	5%
Annual change in benefits	7%
Estimated costs	
Development costs	
Hardware (servers, networking equipment)	RM 10,000
Software development (student project — labour contributed as coursework)	RM 0
Production costs	
Cloud hosting — AWS EC2 + S3	RM 4,200 per year
IS Salaries (post-launch maintenance team)	RM 42,000 per year
Advertisement / Marketing	RM 5,000 per year

Maintenance and updates	RM 3,000 per year
Estimated benefits	
Increased user base (remote/hybrid students, new enrolments)	RM 60,000 per year
Savings (reduced printing, admin, avoided legacy costs)	RM 30,000 per year

Costs	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development cost						
Hardware	11,000	—	—	—	—	—
Software development	0	—	—	—	—	—
Total	11,000	—	—	—	—	—
Production cost						
Cloud hosting	—	4,620	4,851	5,094	5,348	5,616
IS Salaries	—	46,200	48,510	50,936	53,482	56,157
Advertisement / Marketing	—	5,500	5,775	6,064	6,367	6,685
Maintenance	—	3,300	3,465	3,638	3,820	4,011
Annual Production Cost	—	59,620	62,601	65,732	69,017	72,469
(Present Value)	—	*54,200*	*51,737*	*49,384*	*47,131*	*45,009*
Accumulated Costs	—	65,200	116,937	166,321	213,452	258,461

Benefits	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Increase Users	—	54,000	57,780	61,825	66,153	70,784
Savings	—	27,000	28,890	30,912	33,076	35,392
Annual Benefits	—	81,000	86,670	92,737	99,229	106,176
(Present Value)	—	*73,636*	*71,628*	*69,671*	*67,790*	*65,963*
Accumulated Benefits	—	73,636	145,264	214,935	282,725	348,688
Gain or Loss	—	(6,436)	28,327	48,614	69,273	90,227
Profitability Index (PI)	1.35					

5.0 OBJECTIVES

These objectives will show what's our goal for this project in term of the system that will make:

1. **Enhance Learning Accessibility**
   The system aims to enhance accessibility by providing recorded lecture content and educational materials to UTM students, postgraduates, and lecturers. By making recordings available on-demand, learners who are unable to attend session in person can still access the same quality of instruction at their own convenience.

2. **Strengthen Peer Learning Network**
   The system is designed to strengthen peer learning networks within the university. It will facilitate collaborative learning by enabling students, lecturer and postgraduate to share knowledge, engage with academic content, and support one another through a structured digital platform.

3. **Provide Actionable Academic Insight**
   The system will leverage Artificial Intelligence (AI) to analyse academic questions and providing users with actionable insights. This AI-driven analysis will help frequently asked questions and trends in academic performance. It also can show performance analytics for user in term of studies.

4. **Build a Sustainable Knowledge Repository**
   The system will build a knowledge repository, specifically to keep Past Year Question collections and other academic resources. This repository will serve as a long-term reference point for students and researchers, ensuring that valuable academic work is preserved and accessible for future use.

5. **Create a study section for students**
   The system will create a space for good students to share a study guide or study tips section for others to follow. Other than that, a study tools will be added for students to use to improve their study such as pomodoro timer.

6.0 SCOPE OF THE PROJECT

With everything have been discussed, there will be a feature or system that will be developed or falls outside the project overage.

In Scope

The following are included in the project:

Users The system will serve the following user groups:

- UTM Students (Undergraduate)
- Postgraduate Students
- Lecturers

Platform The primary deliverable will be a web application. A mobile application may be developed as an extension.

Key Features

- Smart Classroom Recording System
- Academic Forum and Discussion Board
- AI Study Assistant
- Carry Mark and Performance Analytics
- Student Collaboration Features
- Study Guide section

Out of Scope

The following are explicitly excluded from this project:

- Upcoming class recording features – live or scheduled class recordings are not within the current project scope.
- External Users – only UTM-affiliated users are supported; the system will not be open to members outside the university.
- Hardware for cameras – integration of physical recording hardware exclusively for this system is not part of it.
- Content creation tools – the system will not include tools for creating or editing video content; it focuses on content distribution and access not production.

7.0 PROJECT PLANNING

7.1 MANAGEMENT ORGANISATION CHART

7.2 Work Breakdown Structure

Analysis

1.1- Go through UTM e-learning system and identify existing issues in the system.

1.2- Conduct an interview regarding the e-learning system with the staff of UTM Digital.

1.3- Define system requirements.

1.4- Prepare project proposal documentation.

Design

2.1- System Architectural Design

2.2- Database Design (student records and recordings)

2.3- Web Interface Design

2.4- AI Feature Planning (Study Recommendation AI, Performance Prediction AI, Progress Summary AI)

Development

3.1- Smart Classroom Recording System

3.2- Academic Forum and Discussion Board

3.3- Carry Mark and Performance Analytics

3.4- AI Study Assistant

3.5- Student Collaboration Features

3.6- Focus Session Controller

Testing

4.1- System Testing

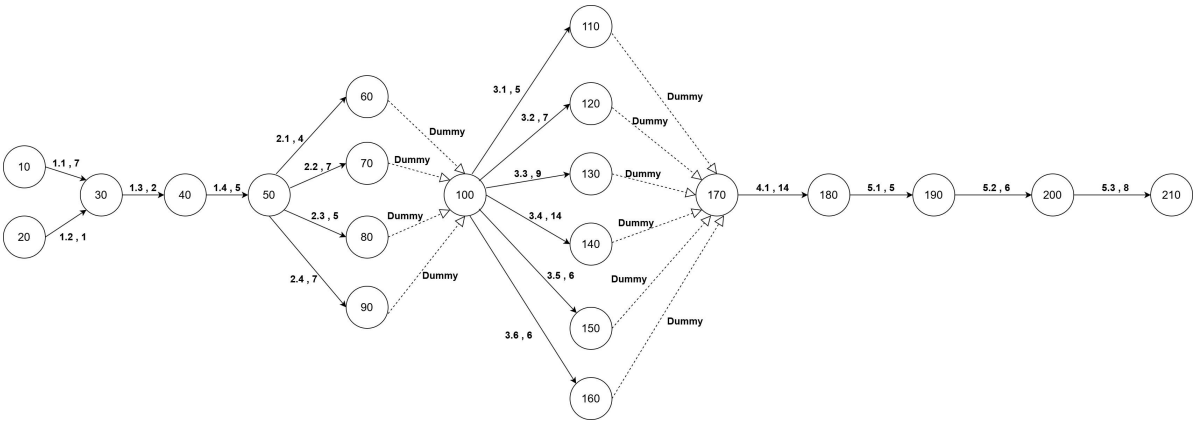
Deployment and Maintenance

5.1- Cloud Hosting Setup

5.2- System Deployment

5.3- Maintenance and Updates

7.3 PERT Chart



Critical Path: 1.1 - 1.3 - 1.4 - 2.2 - 3.4 - 4.1 - 5.1 - 5.2 - 5.3

Length: 7 + 2 + 5 + 7 + 14 + 14 + 5 + 6 + 8 = 68 days

7.4 Gantt Chart

8.0 BENEFIT AND OVERALL SUMMARY OF PROPOSED SYSTEM

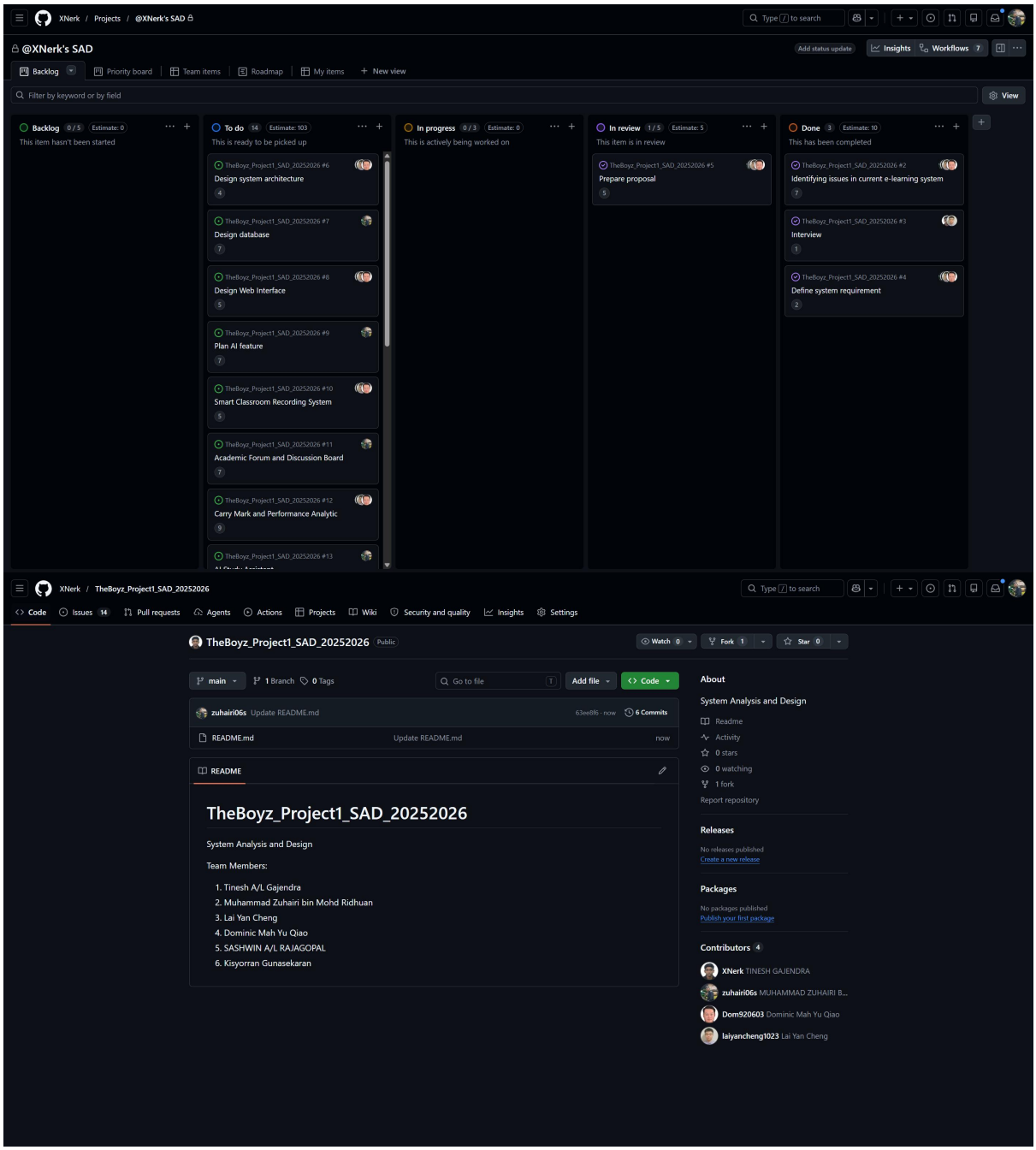
The heart of **Eduhub** is more than just software; it's about creating a supportive, modern space that truly understands the rhythm of university life. By moving past the friction of outdated, manual habits, the platform transforms the academic experience into something that feels fluid and manageable. It lifts the heavy lifting of logistics off the shoulders of students and lecturers, freeing them to spend their mental energy on what matters, meaningful learning and groundbreaking research.

The most immediate impact is a new sense of efficiency. By gathering task lists, document sharing and project tracking into one digital home, the system cuts through the noise, allowing students to accomplish more with less effort. No more digging through endless folders or chat histories. Whether it's a specific research note or a past exam paper, everything is exactly where it should be. This level of organization ensures that hard work is preserved and always within reach, never falling through the cracks of a busy semester.

Collaboration also shifts from a chore to a natural conversation. Rather than feeling isolated islands, students can lean on one another through integrated forums, peer feedback, and direct tutoring sessions. With real-time updates, every team member stays in the loop, eliminating the frustration of redundant work and ensuring the final project is a true reflection of the group's best efforts.

Ultimately, the system acts as a reliable partner in every student's journey. Through intuitive progress trackers and gentle reminders, it provides a clear map of where you stand and where you're headed. Deadlines no longer "sneak up" and grades become a transparent metric for growth rather than a source of anxiety. Because the interface is designed with a human-first approach, it's accessible and easy for anyone to use. Building confidence and peace of mind needed to truly thrive at UTM.

GitHub

GitHub Project link:

https://github.com/XNerk/TheBoyz_Project1_SAD_20252026